

Completing Short Algebraic and Segment Proofs

Answer Key

High-school geometry

Quick Answers

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| 1. (a) $x = 5$ cm, — | 5. (b) the longer distance = 15.5 m, — |
| 2. (a) $x = 7$, (b) the distance from the second post to the third = 11 m | 6. — |
| 3. — | 7. (a) the unknown length = 15 m, — |
| 4. (a) $x = 8$ cm, — | 8. (b) the outer span = 7.3 m, — |
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What is verified

1 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry
HSA-REI.A.1 – *problems 1, 4, 7*
HSG-CO.C – *problems 2, 3, 5, 6, 8*
Difficulty 1–5 of 5 across 14 tagged checks.

1. Shelf support: $3x + 7 = 22$ (cm). (a) Find x . (b) Give the two missing reasons.
(a) Undo the bracket, then undo the tripling:

$$3x + 7 = 22$$

$$3x = 15$$

So the spacer is 5 cm long.

$x = 5$ cm

(b) Line 2: **subtraction property of equality** (7 subtracted from both sides). Line 3: **division property of equality** (both sides divided by 3). Naming the operation without naming the property earns half credit — a proof cites the property.

2. Trail markers A, B, C : $AB = 2x$, $BC = x + 4$, $AC = 25$ (m). (a) Find x . (b) Find BC .

(a) B lies between A and C , so the Segment Addition Postulate gives $AB + BC = AC$:

$$2x + (x + 4) = 25$$

$$3x + 4 = 25$$

$$3x = 21$$

So $x = 7$.

(b) $BC = x + 4 = 7 + 4$. Check: $AB = 2(7) = 14$ m and $14 + 11 = 25$ m.

$x = 7$

$BC = 11$ m

3. Handrail $\overline{PQ}$ with midpoint M : complete the reasons for $PM = \frac{1}{2}PQ$.

Reasons, in order:

2. **Definition of midpoint** — a midpoint divides the segment into two congruent segments.

3. **Definition of congruent segments** — congruent segments have equal lengths, which is what turns the congruence mark into an equation.

4. **Segment Addition Postulate** — M lies between P and Q , so the two pieces sum to the whole.

Line 5 is already given: substituting PM for MQ gives $2PM = PQ$, and dividing by 2 gives the result.

Grading: full credit needs the definition of midpoint and the definition of congruent segments as *separate* reasons — the first gives congruence, the second gives equality of lengths.

4. Cable runs: $5(x - 3) = 2x + 9$ (cm). (a) Find x . (b) Reason for each line.

(a)

$$5(x - 3) = 2x + 9$$

$$5x - 15 = 2x + 9$$

$$3x - 15 = 9$$

$$3x = 24$$

So $x = 8$ cm. Check: $5(8 - 3) = 25$ and $2(8) + 9 = 25$.

$x = 8$ cm

(b) Line 2: **distributive property**. Line 3: **subtraction property of equality** ($2x$ from both sides). Line 4: **addition property of equality** (15 added to both sides). Line 5: **division property of equality**.

5. Markers A , B , C , D in order with $AB = CD$. (a) Prove $AC = BD$. (b) With $AB = CD = 3.5$ m and $BC = 12$ m, find AC .

(a) **Model proof:** $AB = CD$ is given. Adding the shared middle distance BC to both sides gives $AB + BC = BC + CD$ by the addition property of equality. By the Segment Addition Postulate, $AB + BC = AC$ and $BC + CD = BD$. Substituting these two into the previous line gives $AC = BD$.

Grading: the argument must (i) add the shared BC to both sides with a named property, and (ii) use the Segment Addition Postulate to rename the two sums. An answer that only asserts “the extra bits are equal so the totals are equal” with no reasons is partial credit.

(b) $AC = AB + BC = 3.5 + 12.$

$$AC = 15.5 \text{ m}$$

6. Same four markers, now given $AC = BD$: complete the reasons for $AB = CD$.

Reasons, in order:

1. **Given.**

2. **Segment Addition Postulate** — B is between A and C .

3. **Segment Addition Postulate** — C is between B and D .

4. **Substitution** — both sums equal the same quantity by line 1, so they equal each other.

5. **Subtraction property of equality** — BC removed from both sides.

Grading: lines 2 and 3 are the same postulate applied to two different betweenness facts; both must be cited. Line 5 must name subtraction, not “cancel”.

7. Garden bed: $P = 2l + 2w$ with $P = 46$ m, $w = 8$ m. (a) Find l . (b) Name the properties. (a)

$$2l + 2(8) = 46$$

$$2l + 16 = 46$$

$$2l = 30$$

So the length is 15 m. Check: $2(15) + 2(8) = 30 + 16 = 46$ m.

$$l = 15 \text{ m}$$

(b) Replacing P by 46 and w by 8: **substitution**. Removing the 16: **subtraction property of equality**. Dividing by 2: **division property of equality**.

8. Footbridge W, X, Y, Z with $WX = YZ$, $XY = 6.4$ m, $WZ = 21.0$ m. (a) Prove $WY = XZ$. (b) Find WX .

(a) **Model proof:** $WX = YZ$ is given. Adding the shared middle span XY to both sides gives $WX + XY = XY + YZ$ by the addition property of equality. By the Segment Addition Postulate, $WX + XY = WY$ and $XY + YZ = XZ$. Substituting gives $WY = XZ$.

Grading: same two requirements as problem 5 — the shared span added with a named property, and the postulate used to rename each sum.

(b) Let w be the length of one outer span. The three spans make the whole bridge, so

$$w + 6.4 + w = 21.0$$

$$2w + 6.4 = 21.0$$

$$2w = 14.6$$

So each outer span is 7.3 m. Check: $7.3 + 6.4 + 7.3 = 21.0$ m.

$$WX = 7.3 \text{ m}$$